

Section A – Concept Application

Scenario 1: Selecting Compatible Core Hardware for an Office Workstation

1. Selection Process for Each Component

- **Step 1: Choose the Processor (CPU)**
 - **What to pick:** A modern 6-core processor with **Integrated Graphics (iGPU)** (such as an Intel Core i5 with Intel UHD Graphics or an AMD Ryzen 5 with Radeon Graphics).
 - **Why:** Integrated graphics eliminate the added purchase cost, heat, and power consumption of a dedicated graphics card for regular office work.
- **Step 2: Choose the Motherboard**
 - **Socket Match:** Pick a motherboard with the exact same CPU socket as your chosen processor (e.g., **LGA 1700** for Intel 12th/13th/14th Gen, or **AM5** for AMD Ryzen 7000/8000/9000 series).
 - **Chipset & Size:** A business-grade **B-series chipset** (Intel B760 or AMD B650) in a **Micro-ATX** form factor provides the best balance of cost, stability, and desk space.
- **Step 3: Choose the RAM (Memory)**
 - **Generation & Capacity:** Pick memory supported by both your motherboard and CPU (**DDR4 or DDR5**).
 - **Configuration:** Select a **16 GB kit (2 × 8 GB)**. Using two sticks instead of one enables **dual-channel mode**, doubling memory bandwidth for smoother office multitasking across spreadsheets, browser tabs, and video calls.
- **Step 4: Choose the Storage Device**
 - **Form Factor & Capacity:** Select a **500 GB or 1 TB M.2 NVMe SSD** that matches the motherboard's M.2 slot key (M-Key, 2280 size) for fast system boot and responsive file loading.

2. How to Confirm Hardware Compatibility

- **Check the Motherboard's CPU Support List:** Visit the motherboard manufacturer's official support webpage. Confirm that the exact CPU model is

listed as compatible and check the minimum BIOS version required to boot it out of the box.

- **Check the Memory QVL (Qualified Vendor List):** Consult the motherboard's official memory QVL to verify that the specific RAM model number and operating speed have been tested and certified for stable operation.
- **Verify M.2 Interface Specifications:** Confirm that the motherboard's M.2 slot supports the NVMe PCIe generation (e.g., PCIe 4.0 x4) and the 2280 physical length of the SSD.
- **Use Automated System Validation:** Input the selected components into a tool like **PCPartPicker** to run an automated check for physical clearances, socket mismatches, and power load compatibility.

Scenario 2: Storage Comparison & Recommendation for Office Workstations

1. Differences Between HDD, SATA SSD, and NVMe SSD

Feature	Traditional HDD (Hard Disk Drive)	SATA SSD (2.5-inch Solid State Drive)	NVMe M.2 SSD
Technology	Mechanical spinning magnetic disks with moving read/write heads.	Flash memory chips with no moving parts.	High-speed Flash memory chips communicating directly over PCIe lanes.
Data Transfer Speeds	Slow: 100 – 200 MB/s	Moderate: up to 550 MB/s	Ultra-Fast: 3,500 – 7,000 + MB/s
Responsiveness (IOPS / Latency)	Low (75 – 100 IOPS); high mechanical seek latency (≈ 12 ms).	High (≈ 90,000 IOPS); low latency (< 0.1 ms).	Extremely High (500,000 + IOPS); near-instant response (≈ 0.02 ms).
Physical Design & Noise	Heavy, sensitive to drops/vibration, audible clicking/spinning noise.	Silent, durable plastic/metal case connected via SATA cables.	Silent, installs directly onto the motherboard without any cables.

2. Recommendation & Technical Justification

- **Recommended Option: M.2 NVMe SSD (500 GB or 1 TB)**
- **Technical Justification:**
 - **Eliminates Daily Multitasking Lag:** Employees running office workflows continuously switch between web browser tabs, accounting/ERP portals, large PDF files, and Microsoft Office spreadsheets. The high random IOPS (Input/Output Operations Per Second) of an NVMe drive ensures applications launch instantly with zero lag.
 - **Fast Boot & Background Task Handling:** Workstations boot into Windows in under 10 seconds. Antivirus scans, software updates, and large file saves run in the background without causing 100% disk usage freezes (a frequent problem with traditional HDDs).
 - **High Reliability & Zero Cables:** Because NVMe drives have no moving mechanical parts, they cannot suffer mechanical head crashes or vibration failures. They screw directly into the motherboard, eliminating data and power cables for a cleaner build with improved airflow.
 - **Cost-Effectiveness:** Mainstream PCIe 3.0 and 4.0 NVMe drives now cost almost the same as older SATA SSDs while delivering up to 7 times the speed, making them the best long-term investment for office productivity.

Scenario 3: Workstation for Sustained Graphics-Intensive Design Software

1. Role of the GPU (Graphics Processing Unit):

- **Hardware Acceleration & Offloading:** Design applications (such as CAD, Photoshop, Illustrator, and 3D modeling) rely on the GPU to process vector mathematics, render textures, and display viewport transformations. This offloads graphical workloads from the CPU, keeping the operating system responsive.
- **Dedicated Video Memory (VRAM):** The GPU's high-speed onboard memory stores large image assets, frame buffers, and rendering textures, preventing system RAM saturation and eliminating render bottlenecks.

2. Role of the Cooling System:

- **Preventing Thermal Throttling:** Running heavy workloads continuously generates high heat. When internal sensors detect excessive temperatures (typically above 80°C–85°C), the GPU and CPU automatically drop their clock speeds to avoid damage, causing software slowdowns and rendering stutter. Efficient cooling keeps clock speeds at their rated boost frequencies.
- **Long-Term System Stability:** Continuous high heat degrades silicon and voltage regulation modules (VRMs). A balanced thermal setup—using effective heatsinks, quality thermal paste, and structured intake/exhaust case airflow—prevents application crashes, system freezes, and unexpected reboots during multi-hour rendering jobs.

Senario 4: Assembling a PC from Components Sourced Across Multiple Vendors

1. Why Hardware Compatibility Matters:

- PC components follow standardized architectural, electrical, and dimensional standards. Sourcing parts from different vendors requires verifying that physical interfaces, communication buses, and electrical requirements align; otherwise, parts will either not fit physically or fail to operate logically.

2. Potential Issues from Incompatible Components:

- **Physical Damage & Socket Mismatch:** Attempting to force an incompatible CPU into a different motherboard socket (e.g., placing an Intel CPU into an AMD AM5 socket) will bend or break fragile socket pins.
- **No-Boot / Memory Failure:** Installing the wrong RAM standard (e.g., DDR5 memory in a DDR4 slot) or unsupported speeds/voltages will trigger POST boot failure and memory detection errors.
- **Insufficient Electrical Power:** An under-rated power supply or one lacking the necessary dedicated 8-pin PCIe power cables will cause instant shutdowns under load or damage components through electrical instability.
- **BIOS / Firmware Incompatibility:** Even if the socket matches, an older motherboard chipset might not recognize a newer-generation processor out of the box without a firmware update, resulting in a blank screen on power-up.
- **Chassis Clearance Bottlenecks:** Sourcing a large triple-fan GPU or a tall tower cooler that exceeds the physical interior clearance of a compact case prevents the case panels from closing.

Scenario 5: Power Supply Unit (PSU) Sizing, Connections, and Pre-Boot Checks

1. How to Choose an Appropriately Rated PSU:

- **Calculate Total System TDP (Thermal Design Power):** Sum the maximum wattage drawn by each component under 100% load:
 - CPU consumption (e.g., 65W–150W)
 - Dedicated GPU consumption (e.g., 100W–250W)
 - Motherboard, RAM, storage drives, and fans ($\approx$ 50W–75W)
- **Apply a 25%–30% Overhead Headroom:** Never size a PSU to run at 100% capacity. Adding a safety buffer accommodates instantaneous transient power spikes, ensures the PSU operates in its peak electrical efficiency curve (typically 50%–70% load), and leaves room for future upgrades.
- **Select an 80 PLUS Certified Unit:** Choose at least an **80 PLUS Bronze or Gold** rating from a reputable tier to ensure clean voltage regulation and built-in protections:
 - OVP (Over Voltage Protection)
 - OCP (Over Current Protection)
 - SCP (Short Circuit Protection)

2. Safety & Connection Checks Before First Boot:

- **Main Motherboard Power (24-Pin ATX):** Confirm the wide 24-pin cable is inserted completely into the motherboard header and the plastic retaining clip has clicked shut.
- **CPU EPS Power (4+4 or 8-Pin):** Ensure the dedicated 8-pin CPU power cable is connected to the top-left of the motherboard (do not confuse this with a PCIe cable).
- **GPU PCIe Power (6+2 Pin):** Ensure the graphics card receives its dedicated 8-pin power cables directly from the PSU; avoid daisy-chaining a single cable for high-power cards.
- **Storage & Peripheral Power:** Ensure SATA power cables are seated firmly in all SATA SSDs/HDDs.
- **Physical Safety Checks:**
 - Verify that motherboard standoffs are installed correctly underneath to prevent the board from shorting against the metal chassis.
 - Check that no loose screws, zip ties, or stray copper wires are resting on the motherboard or GPU PCB.
 - Ensure all internal cables are clear of spinning cooling fan blades.
 - Verify the PSU voltage selector switch (if present) is set to your regional mains voltage (230V or 115V) before connecting the wall cable.

Scenario 6: Handover Verification, POST Validation, and Troubleshooting

1. Checks to Confirm the System Passes POST Successfully:

- **Single Short Beep (Audio Check):** If a motherboard chassis speaker is connected, a single short beep indicates successful Power-On Self-Test (POST) completion.
- **Motherboard Debug LEDs / Display:** Modern motherboards feature four diagnostic LEDs (**CPU**, **DRAM**, **VGA**, **BOOT**). Successful POST cycles through each light and shuts them off, or ends with the green **BOOT** light lit. On boards with a 2-digit hex display, it settles on standard codes (such as A0 or AA).
- **Visual Video Output (POST Screen / BIOS):** The monitor wakes up from standby and displays the motherboard manufacturer logo, system specifications, and prompts to press **DEL** or **F2** to enter setup.
- **Hardware Detection Inside BIOS:** Enter BIOS and verify that:
 - All installed RAM capacity and channels (e.g., 16 GB or 32 GB in Dual-Channel) are detected.
 - Primary storage drives (M.2 NVMe SSD) appear with their full reported capacity.
 - CPU and system temperatures are operating within safe idle ranges (typically 30°C–45°C).

2. What to Do If the System Fails POST:

- **Observe the Diagnostic Clues:** Look at the motherboard's debug LEDs or listen to beep codes:
 - *DRAM light on / Repeating beeps:* RAM memory detection failure.
 - *VGA light on / 1 long + 2 short beeps:* Graphics card not detected or video cable disconnected.
 - *CPU light on:* CPU seated improperly, bent socket pins, or missing 8-pin EPS CPU power.
- **Reseat Core Components:** Power down, unplug the machine, and firmly reseat:
 1. The RAM modules into the primary slots (DIMM A2/B2).
 2. The GPU into the top PCIe x16 slot.
 3. All PSU power connectors.
- **Perform Single-Stick RAM Isolation:** If memory issues persist, remove all RAM modules except one stick in Slot A2 to identify whether a specific stick or slot is defective.
- **Reset CMOS Settings:** Disconnect power, remove the motherboard CR2032 coin-cell battery for 2 minutes (or short the CLRTC jumper pins), and reinstall to clear corrupted BIOS settings and reset hardware timing tables to factory defaults.